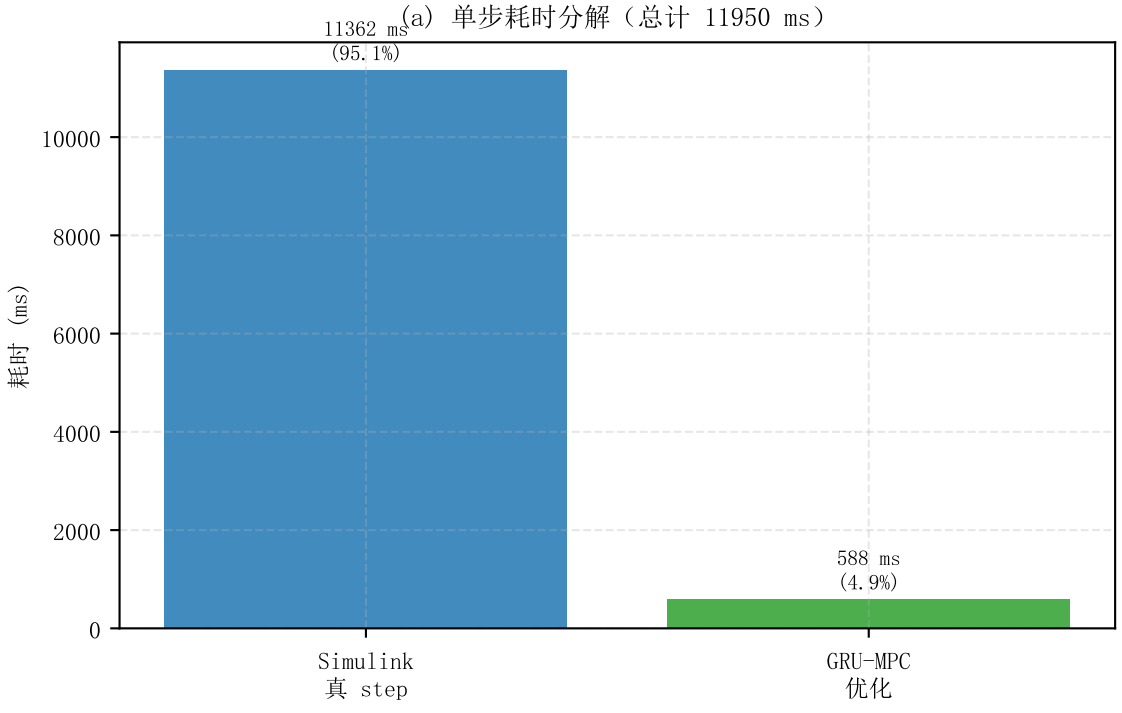
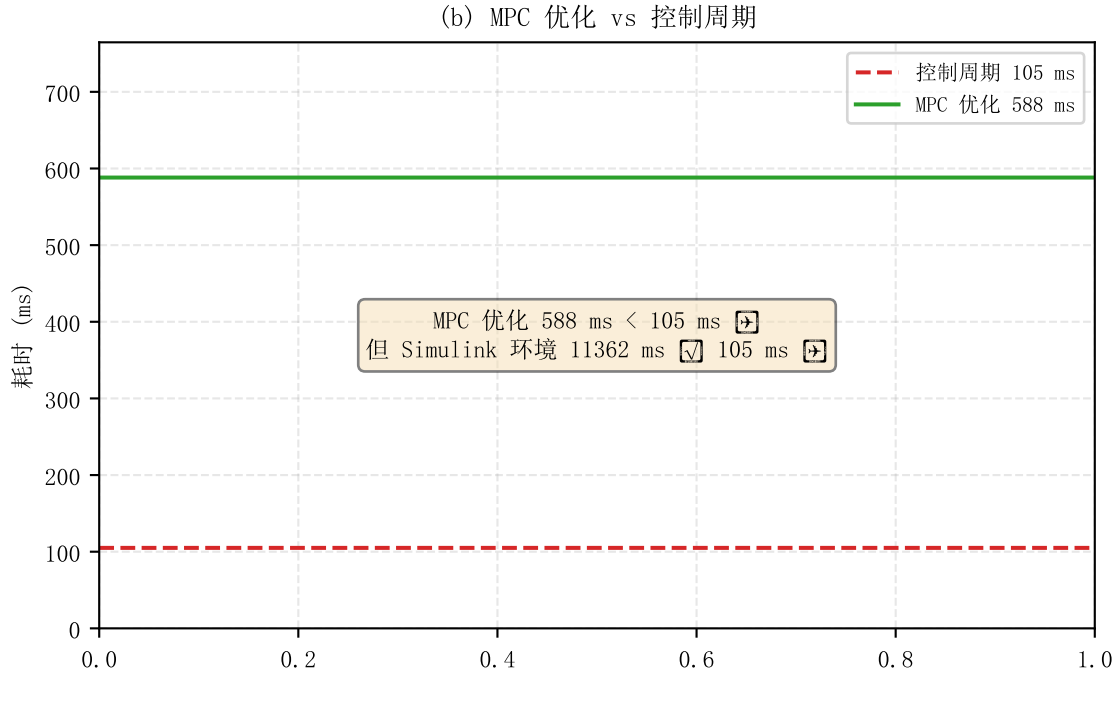
图 D　计算开销分析


(a) 单步耗时分解（总计 11950 ms）
11362 ms
(95.1%)
588 ms
(4.9%)
耗时 (ms)
0
2000
4000
6000
8000
10000
Simulink
真 step
GRU-MPC
优化


(b) MPC 优化 vs 控制周期
控制周期 105 ms
MPC 优化 588 ms
MPC 优化 588 ms < 105 ms
但 Simulink 环境 11362 ms > 105 ms
耗时 (ms)
0
100
200
300
400
500
600
700
0.0
0.2
0.4
0.6
0.8
1.0